Utthan Singh Roy

Roll No.:23035010081
Online BSc (Hons.) - Data Science and AI
Indian Institute of Technology, Guwahati

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Education

Degree | Discipline	Institute/Board	CGPA/Percentage	Year
BSc (Hons.) Data Science and AI	Indian Institute of Technology, Guwahati	8.34 (Current)	2023-Present
B.Tech Electrical Engineering	Madan Mohan Malaviya University of Technology, Gorakhpur	8.36 (Current)	2023-Present
Senior Secondary	CBSE Board, St. Joseph's School, Shaktinagar	94.5%	2021
Secondary	CBSE Board, St. Joseph's School, Shaktinagar	93.8%	2019

Experience

- **Indian Institute of Technology (IIT) Guwahati** *Jul. 2026 – Present*
  LLM Application Engineering Intern Guwahati, Assam
  - Architected a low-latency Hybrid RAG pipeline combining ChromaDB/BM25 retrieval with asynchronous LCEL token streaming to minimize Time-to-First-Token (TTFT).
  - Engineered citation and deterministic fallback guardrails for LLM outputs, improving reliability of production-oriented microservice responses.

Projects

- **AI-Powered Market Regime Shift Classifier** *Feb. 2026 – Jun. 2026*
  Python, XGBoost, FinBERT, Streamlit, GitHub Actions GitHub
  - Engineered a cost-sensitive XGBoost model on 41 quantitative and FinBERT sentiment features, achieving 0.97 ROC-AUC for systemic market stress detection.
  - Built a real-time inference dashboard and automated feature engineering, deployment, and model updates through GitHub Actions CI/CD.
- **Physics-Validated AI Architecture for Autonomous Grid Resilience** *Apr. 2026 – Jul. 2026*
  Python, LightGBM, SAC Reinforcement Learning, LangGraph GitHub
  - Designed Soft Actor-Critic (SAC) reinforcement learning policies with a LightGBM forecasting layer achieving 1.43% MAPE and 100% convergence in AC power-flow simulations.
  - Implemented a deterministic validation firewall for LLM-generated targets, dynamically rejecting invalid outputs and improving the Grid Resilience Index by 6.41 points.
- **Automated Sustainable Portfolio Optimizer** *Oct. 2025 – Jan. 2026*
  Python, XGBoost, NSGA-II, GitHub Actions, Pandas GitHub
  - Combined NSGA-II optimization with XGBoost to forecast 30-day Nifty 50 directionality and dynamically optimize asset weights under tail-risk constraints (CVaR).
  - Automated daily data ingestion, model retraining, and deployment through a GitHub Actions pipeline.

Technical Skills

- **Programming**: Python, SQL, C, Java*, R*
- **AI/ML**: PyTorch, TensorFlow, Scikit-Learn, XGBoost, LightGBM, SAC, NSGA-II, FinBERT, SHAP*
- **LLM/Agentic AI**: LangGraph, RAG, LCEL, ChromaDB, BM25, API Integration
- **MLOps/Tools**: Git, GitHub Actions, Streamlit, SQLite, Pandas, NumPy ** Elementary proficiency*

Key Courses Taken

- **Computer Science & AI**: Data Structures, Algorithm Design & Analysis, Deep Learning for Computer Vision, Hardware-Aware Deep Learning, LLM Application Engineering
- **Mathematics & Analytics**: Linear Algebra, Optimization, Statistical Inferencing, Time Series Analysis
- **Systems & Engineering**: Control System Engineering, Digital Control System, Microprocessors

Positions of Responsibility

- **Volunteer**, AI Confluence, IIT Guwahati *2024*

Certifications & Achievements

- **Explainable AI (XAI) Specialization**, Coursera, Duke University *2026*
- **AI in Healthcare Specialization**, Coursera, Stanford University *2026*
- **Qualified GATE 2026**, Data Science & Artificial Intelligence (DA) *2026*